

Tuning the electronic heat capacity and thermal Schottky anomaly of monolayer β_{12} -borophene via adsorbed gas molecules

Le T.T. Phuong^a, Tran Cong Phong^{b,c,*}

^a Faculty of Physics, University of Education, Hue University, Hue, 530000, Vietnam

^b Atomic Molecular and Optical Physics Research Group, Institute for Advanced Study in Technology, Ton Duc Thang University, Ho Chi Minh City, Vietnam

^c Faculty of Electrical and Electronics Engineering, Ton Duc Thang University, Ho Chi Minh City, Vietnam

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ABSTRACT

Owing to the inherent inversion symmetry between five atoms $\{a, b, c, d, e\}$ and metallic phase, β_{12} -borophene possess fantastic physical properties, such as various electronic phase transitions in the presence of external perturbations. Here, the effect of adsorbed gas molecules (NH_3 , NO , NO_2 , and CO) on the electronic heat capacity (EHC) and thermal Schottky anomaly of β_{12} -borophene is calculated employing the tight-binding model and the Green's function approach. The results show that the gas-adsorbed EHC is enhanced with the dominant contribution of NO on individual $\{a, b, d, e\}$ atoms, while it is NO_2 on c atoms. However, EHC is suppressed when the molecule is adsorbed on the pair of boron atoms. Moreover, the enhancement of EHC is dominated by the simultaneous adsorption of NH_3 on all atoms. On average, the pristine Schottky anomaly moves to the higher thermal energies upon the adsorption of gas molecules. These distinct differences originate from the generation of various new energy levels. This study provides a detailed understanding of β_{12} -borophene response to the gas molecules for tuning field effect transistor-based gas sensors.

1. Introduction

Nowadays, beginning with graphene [1,2], two-dimensional (2D) materials such as silicene [3,4], germanene [5], stanene [6], molybdenum disulfide [7], and phosphorene [8,9], provide amazing physical and chemical properties compared to bulk materials. Accordingly, a wide variety of applications is proposed for them in different parts of the industry. In contrast to graphene with massless Dirac fermions, other gapped systems (with massive fermions) draw the industry's attention to consider them, e.g., in field-effect transistors due to their high on-off current ratio [10,11].

In addition to the long list of present 2D active materials nowadays, β_{12} -borophene has attracted researchers due to its Dirac-triplet quasiparticles [12–14]. This implies that β_{12} -borophene hosts two different types of charge carriers simultaneously, which is a remarkable point for the electronic properties of materials such as electronic transport. β_{12} -borophene is the most stable allotrope among striped, β_{12} , χ_3 , and honeycomb ones [15,16]. Independent of how boron atoms are arranged to form the borophene sheet, various metallic phases have been found for borophene [17–25]. This, in turn, results in various electronic band structures.

Turning to the practical applications of 2D materials, one can think of the electric circuits including different segments such as memories, diodes, capacitors, transistors, and highly sensitive sensors [26–28]. For instance, compared to traditional sensors, graphene-based sensors [29–31] have provided faster response and recovery time. It has also been found that the intrinsic properties of borophene-based sensors can effectively be tuned via hazardous gases [32–35] such that the adsorption energy is high. Moreover, it has been suggested that the adsorption of gas molecules such as H_2O , NH_3 , CO , NO_2 , and NO , can tune the performance and electronic properties of 2D materials-based sensors [36–38].

Although electronic properties of gas sensing in β_{12} -borophene have been studied in various ways [32–35], to the best of our knowledge, thermal properties have not been addressed so far in the literature. In this work, we focus on this matter to see how the thermal properties of β_{12} -borophene can be affected via gas molecules (GMs). Among various thermal quantities, we proceed with the electronic heat capacity, which defines as the ratio of the heat added to/removed from the matter to the resulting temperature change in the system. Since GMs tune the electronic features of the system, one might study the effect of adsorbed

* Corresponding author at: Atomic Molecular and Optical Physics Research Group, Institute for Advanced Study in Technology, Ton Duc Thang University, Ho Chi Minh City, Vietnam.

E-mail addresses: ltphuong@hueuni.edu.vn (L.T.T. Phuong), tranconghong@tdtu.edu.vn (T.C. Phong).

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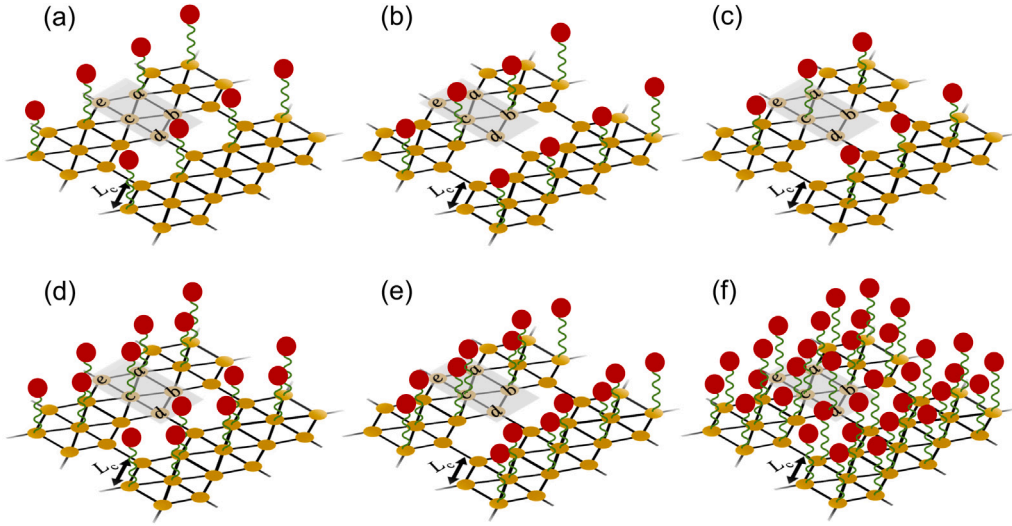


Fig. 1. Atomic structure of β_{12} -borophene including five boron atoms (light brown spheres) $\{a, b, c, d, e\}$ in its unit cell with the lattice constant of $L_c \approx 2.926 \text{ \AA}$ [12–14] in the presence of adsorbed gas molecules (red spheres) on (a) a , (b) b , (c) c , (d) $\{a, c\}$, (e) $\{b, c\}$, and (f) all atoms.

GMs on the heat capacity of β_{12} -borophene. In doing so, we employ the tight-binding model and Green's function technique.

The rest of the paper is organized as follows. In Section 2, we focus on the effective Hamiltonian model of β_{12} -borophene in the absence and presence of GMs (NH_3 , NO , NO_2 , and CO). In Section 3, we turn to the electronic heat capacity of β_{12} -borophene. In Section 4, we present the numerical results and finally, we provide a summary of findings in Section 5.

2. Tight-binding Hamiltonian model

To study the electronic heat capacity of β_{12} -borophene, the behavior of the electronic density of states (DOS) is important to be tracked. First, a Hamiltonian model should be built for the energy levels in DOS. It is well known that the major contribution of orbitals in a graphene lattice originates from the p_z orbitals. Due to similarities between the atomic structure of graphene and β_{12} -borophene, one expects dominant role of p_z orbitals of boron atoms in determining the electronic properties of β_{12} -borophene [12–14]. Thus, for the unit cell of β_{12} -borophene with five atoms $\{a, b, c, d, e\}$ and the lattice constant of $L_c \approx 2.926 \text{ \AA}$ [12–14], see Fig. 1, the following tight-binding Hamiltonian can be proposed; $H_0 = \sum_{\ell} \epsilon_{\ell} c_{\ell}^{\dagger} c_{\ell} + \sum_{\ell k} t_{\ell k} c_{\ell}^{\dagger} c_k + \text{H.c.}$, where $c_{\ell}^{\dagger}$ (c_{ℓ}) creates (annihilates) an electron at site ℓ .

In the absence of GM, we have a 5×5 matrix for the Hamiltonian, given by [12]

$$H_0(\vec{k}) = \begin{pmatrix} \epsilon_a & h_{12} & h_{13} & 0 & h_{15} \\ h_{12}^* & \epsilon_b & \frac{t_{bc}}{t_{ab}} h_{12} & t_{bd} f_{\vec{k}}^* & 0 \\ h_{13}^* & \frac{t_{bc}}{t_{ab}} h_{12}^* & \epsilon_c & \frac{t_{cd}}{t_{ab}} h_{12} & t_{ce} f_{\vec{k}}^* \\ 0 & t_{bd} f_{\vec{k}} & \frac{t_{cd}}{t_{ab}} h_{12}^* & \epsilon_d & \frac{t_{de}}{t_{ab}} h_{12} \\ h_{15}^* & 0 & t_{ce} f_{\vec{k}} & \frac{t_{de}}{t_{ab}} h_{12}^* & \epsilon_e \end{pmatrix}, \quad (1)$$

where $\vec{k} = (k_x, k_y)$ is the momentum, $h_{12} = 2t_{ab}f_{\vec{k}} \cos(k_x L_c/2)$, $h_{13} = t_{ac}f_{\vec{k}}^*$, $h_{15} = t_{ae}f_{\vec{k}}$, $t_{\ell k}$ is the hopping energy, ϵ_{ℓ} is the on-site energy, and $f_{\vec{k}} = \exp[ik_y L_c/2\sqrt{3}]$ [12,13]. For the hopping parameters, we use the inversion symmetry between boron atoms in β_{12} -borophene [13, 39], leading to $\epsilon_a = \epsilon_e = +0.196$, $\epsilon_b = \epsilon_d = -0.058$, $\epsilon_c = -0.845$, $t_{ab} = t_{de} = -2.04$, $t_{ac} = t_{ce} = -1.79$, and $t_{bc} = t_{cd} = -1.84$, $t_{bd} = -1.91$, and $t_{ae} = -2.12$ (all in units of eV).

In the presence of GM, we proceed with the overlap of s orbitals from the GM with p_z orbitals of boron atoms to modulate the electronic properties of β_{12} -borophene, resulting in a 6×6 matrix for the Hamiltonian, given by

$$H(\vec{k}) = \begin{pmatrix} \epsilon_a & h_{12} & h_{13} & 0 & h_{15} & \tilde{t}_a \\ h_{12}^* & \epsilon_b & \frac{t_{bc}}{t_{ab}} h_{12} & t_{bd} f_{\vec{k}}^* & 0 & \tilde{t}_b \\ h_{13}^* & \frac{t_{bc}}{t_{ab}} h_{12}^* & \epsilon_c & \frac{t_{cd}}{t_{ab}} h_{12} & t_{ce} f_{\vec{k}}^* & \tilde{t}_c \\ 0 & t_{bd} f_{\vec{k}} & \frac{t_{cd}}{t_{ab}} h_{12}^* & \epsilon_d & \frac{t_{de}}{t_{ab}} h_{12} & \tilde{t}_d \\ h_{15}^* & 0 & t_{ce} f_{\vec{k}} & \frac{t_{de}}{t_{ab}} h_{12}^* & \epsilon_e & \tilde{t}_e \\ \tilde{t}_a & \tilde{t}_b & \tilde{t}_c & \tilde{t}_d & \tilde{t}_e & \tilde{\epsilon} \end{pmatrix}, \quad (2)$$

where $\tilde{t}_{\ell} = \tilde{t}_{\ell} \frac{L_c}{d}$ denotes the hopping energy induced by GM [40–42], in which $\tilde{t}_{\ell}$ is the average of other hopping energies on atom ℓ and d is the distance between GM and borophene sheet. Due to the interaction between the gas molecules and the host boron atoms, the boron atoms are slightly shaken (like the presence of strain) compared to their pristine position. As a result of this, a pseudomagnetic field originating from such interaction is induced in the host system, producing a vector potential for changing the propagation direction of atomic waves. This is that of Harrison's rule, which mainly deals with the wavefunction of particles [42]. Thus, the hopping energies from one site to the other site are modulated because of the crosstalk between gas molecules and host boron atoms. Accordingly, we have a linear modulation of hopping energies within the framework of linear expansion of the Harrison rule. As the first approximation, we include this effect as the role of an extra potential in the Hamiltonian, last row and column in the matrix representation, through the averaging process for $\tilde{t}_{\ell}$, while neglecting the modulations of original pristine Hamiltonian. Moreover, $\tilde{\epsilon}$ is the on-site energy induced by GM. For NH_3 , NO , NO_2 , and CO , we respectively implement $d = 1.63, 1.38, 1.57, 1.48$ (all in units of $\text{\AA}$) and $\tilde{\epsilon} = 1.11, 0.95, 1.75$, and 1.19 (all in units of eV) [40].

3. Electronic heat capacity

The next step to get closer to the electronic heat capacity [43], the main target quantity in this paper. We first calculate the electronic DOS of β_{12} -borophene in the absence and presence of GM [44]. In

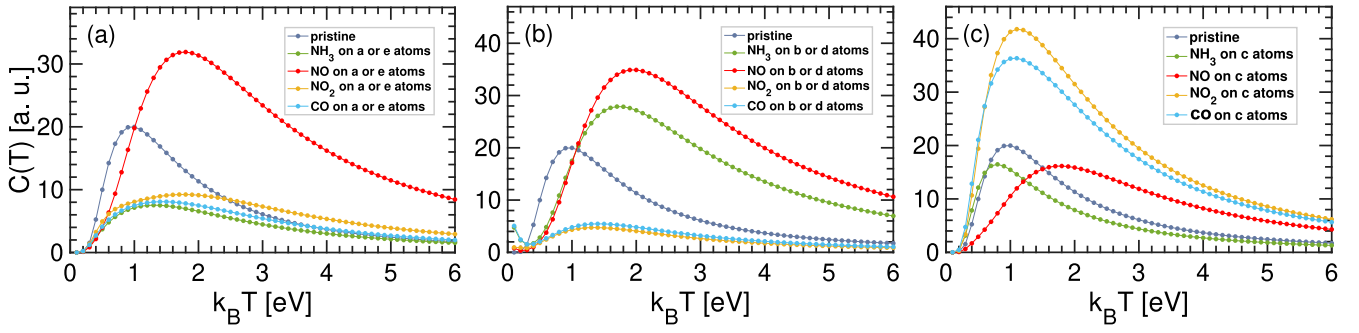


Fig. 2. The electronic heat capacity of β_{12} -borophene in the absence and presence of GMs (NH_3 , NO , NO_2 , and CO); (a) GMs on a/e atoms, (b) GMs on b/d atoms, and (c) GMs on c atoms. The EHC is enhanced with the gas molecules with the dominant contribution of NO on individual $\{a, b, d, e\}$ atoms, while it is NO_2 on c atoms. The Schottky anomaly is shifted to the higher thermal energies with the GMs adsorbed on individual boron atoms.

doing so, we employ the Green's function approach with the aid of the Hamiltonian in Eq. (2):

$$G(\vec{k}, \mathcal{E}) = \frac{1}{\mathcal{E} + i\eta - H(\vec{k})}, \quad (3)$$

where $\eta = 2$ meV is a phenomenological energy factor that tunes the sharpness of singularities in Green's functions. The electronic DOS can be calculated via [45–48]:

$$D(\mathcal{E}) = -\frac{1}{\pi} \text{Tr} \sum_{\vec{k}} \text{Im} G(\vec{k}, \mathcal{E}). \quad (4)$$

Therefore, the adsorption effect of GM as well as the propagation of boron's electrons on a specific boron atom of lattice is included in the diagonal elements of the Green's function matrix.

Afterwards, we proceed to calculate the electronic heat capacity (EHC). EHC is usually defined as the ratio of the heat added to (or removed from) the matter to the resulting temperature change. Generally, there exist two phononic and electronic mechanisms for the thermal behavior of the matter in response to the temperature change, but in the cases of high electron concentrations in most solid state materials, the electronic role is dominant. Thus, EHC basically reflects the details of the excitation spectrum [49]. For this purpose, we first obtain the energy change for finite temperatures:

$$\Delta U = \int_0^\infty d\mathcal{E} \mathcal{E} D(\mathcal{E}) f(\mathcal{E}, T) - \int_0^{\mathcal{E}_F} d\mathcal{E} \mathcal{E} D(\mathcal{E}), \quad (5)$$

in which $f(\mathcal{E}, T) = 1/[\exp(\mathcal{E}/k_B T) + 1]$ (k_B is the Boltzmann constant) is the Fermi-Dirac distribution function and $D(\mathcal{E})$ is the electronic DOS given by Eq. (4). The total number of the electrons $N = \int_0^{\mathcal{E}_F} d\mathcal{E} D(\mathcal{E})$ would help to rewrite the above equation as $\Delta U = \left(\int_0^{\mathcal{E}_F} + \int_{\mathcal{E}_F}^\infty \right) d\mathcal{E} (\mathcal{E} - \mathcal{E}_F) D(\mathcal{E}) f(\mathcal{E}, T) - \int_0^{\mathcal{E}_F} d\mathcal{E} (\mathcal{E} - \mathcal{E}_F) D(\mathcal{E}) = \int_0^{\mathcal{E}_F} d\mathcal{E} (\mathcal{E} - \mathcal{E}_F) D(\mathcal{E}) [f(\mathcal{E}, T) - 1] + \int_{\mathcal{E}_F}^\infty d\mathcal{E} (\mathcal{E} - \mathcal{E}_F) f(\mathcal{E}, T)$. Accordingly, derivation with respect to the temperature at $\mathcal{E}_F = 0$ as the origin of energy results in the following solution for the EHC [50–57]:

$$C(T) = \int_{-\infty}^{+\infty} \mathcal{E} D(\mathcal{E}) \frac{\partial f(\mathcal{E}, T)}{\partial T} d\mathcal{E}. \quad (6)$$

4. Results

From Fig. 1 and Eq. (2), the position of GM on the host boron atoms becomes important so various electronic and thermal responses are expected to appear. From the hopping energies described before based on the inversion symmetry between boron atoms of β_{12} -borophene, the same responses for $\{a$ and $e\}$ or $\{b$ and $d\}$ atoms are expected to be found. In what follows, we analyze three mechanisms when the GM is adsorbed on individual $\{a/e, b/d, c\}$ atoms, on pair $\{a/e, c\}$ or $\{b/d, c\}$ atoms, and on all $\{a, b, c, d, e\}$ atoms. The main origin of all following changes is the modulation of energy levels in the electronic band structure, which can be clearly detected in the electronic DOS [58].

For the pristine tight-binding Hamiltonian, i.e., in the absence of GMs, we plot the temperature-dependent EHC of β_{12} -borophene in the dark blue lines of all panels in Figs. 2–5. It can be seen that EHC has two inversed responses at low and high temperatures associated with critical thermal energy $k_B T_c \simeq 1$ eV (the Schottky anomaly). It is called anomalous because the heat capacity usually increases with temperature, or stays constant [43], while the heat capacity of a solid at low temperature has a peak. This critical temperature can be understood from the fact that the probability of transition increases with the temperature because of quantum effects at low temperatures up to T_c at which the states are filled. Above T_c , EHC starts to decrease with temperature due to the decrease in the probability of transition and thermal effects at high temperatures.

In the presence of GMs on a/e atoms, Fig. 2(a), the Schottky anomaly shifts to the higher energies, namely to 1.3 eV, 1.8 eV, and 2 eV, respectively, with the adsorption of NH_3/CO , NO , and NO_2 . However, at the same time, although the intensity of EHC decreases with adsorbed NH_3 , CO , and NO_2 , it increases with the adsorption of NO . This is simply because of the smallest distance between NO and β_{12} -borophene sheet among all molecules, as well as the smallest induced on-site energy, which in turn, affects the contribution of hopping energies in the electronic properties of β_{12} -borophene. As the temperature is increased, thermal effects are dominated and all responses fall on top of each other independent of the GM.

Within the same manner, in the presence of GMs on b/d atoms, Fig. 2(b), the EHC intensity decreases with the adsorption of NO_2 and CO molecules, while it increases with NH_3 and NO ones. We still find the substantial contribution from NO for the same reasons above-mentioned, however, the same hopping energies between $\{a, b\}$ and $\{d, e\}$ atoms due to the inversion symmetry feature is another reason for this result. Accordingly, the pristine Schottky anomaly at 1 eV shifts to 1.5 eV, 1.7 eV, and 2 eV, respectively, with the adsorption of NO_2/CO , NH_3 , and NO . Interestingly, there is a nontrivial phenomenon at low temperatures for which the EHC in the presence of GMs starts from non-zero intensities rather than expected zero intensity. Our analysis shows that this might be because of on-site energies of the pristine model for $\{b$ and $d\}$ atoms which are the smallest ones compared to all other on-site energies. This implies that the probability of transition from the lowest states of energy in the system at low temperatures is still high in contrast to the on-site energies of the pristine model for $\{a$ and $e\}$ atoms.

Next, in Fig. 2(c), we investigate the adsorption of GMs on the central c atoms in the honeycomb lattice of β_{12} -borophene. Due to the inversion symmetry feature of our model, the wavefunctions in the vicinity of c atoms cancel each other (meaning that the hopping energies are not important in this configuration), and a different response is expected compared to the two previous configurations. In this situation, thus, only the on-site energies play a vital role in determining the EHC intensity and the Schottky anomaly position. Among four GMs, the on-site energy of NO_2 (CO) is the first (second) strongest one, therefore, the

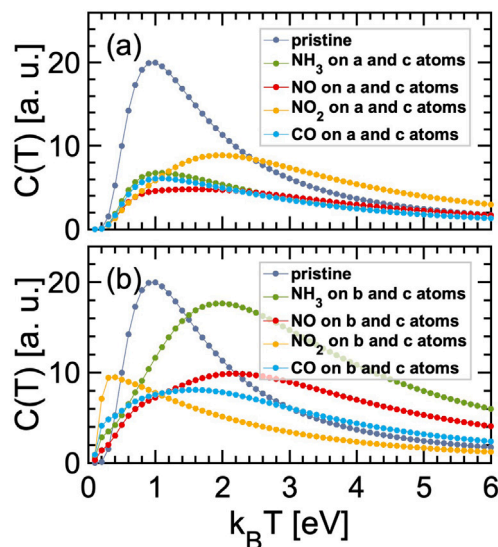


Fig. 3. The same as Fig. 2 but for the adsorption of GMs on (a) $\{a/e \text{ and } c\}$ atoms and (b) $\{b/d \text{ and } c\}$ atoms. EHC is suppressed when GMs are adsorbed on pair of atoms associated with the shift of the Schottky anomaly.

EHC intensity is enhanced for these molecules. Subsequently, the EHC intensity decreases for two other molecules. Although the perturbed Schottky anomaly still moves to higher thermal energies, namely to 1.1 eV, 1.2 eV, and 1.8 eV respectively with the adsorption of NO, CO, and NO₂, it moves to lower thermal energy of 0.7 eV when NH₃ is adsorbed on c atoms. This is because of its longer distance from the borophene sheet.

The next configuration for the adsorption of GMs on the boron atoms to tune the electronic and thermal properties of β_{12} -borophene is the adsorption on pair of boron atoms. To understand the effect of different GMs on the EHC and the Schottky anomaly, we give, in Fig. 3, the EHC as a function of temperature. Regardless of the GM, upon adsorption on $\{a/e \text{ and } c\}$ atoms, it is observed, in Fig. 3(a), that EHC is suppressed by a factor of 2 and 4, respectively, for the adsorption of NO₂ and NH₃/NO/CO. The position of the Schottky anomaly remains the same for NH₃/CO, while it is shifted from pristine 1 eV to 2 eV for NO₂/NO. Turning to the adsorption of GMs on $\{b/d \text{ and } c\}$ atoms in Fig. 3(b), we still observe a suppression of EHC but weaker than the previous configuration, while the Schottky anomaly behaves differently; for the adsorption of NO₂, strong hopping parameter built by coupling of NO₂ and boron atoms as well as the weak on-site energy of b/d atoms in the pristine model result in a Schottky anomaly at low thermal energy of 0.3 eV. On the other hand, the Schottky anomaly of β_{12} -borophene in the presence of other three GMs (NH₃, NO, and CO) moves to the higher energies of 2 eV, 2 eV, and 1.5 eV, respectively.

To further understand the model as well as to provide a comprehensive study for the adsorption of GM on β_{12} -borophene, we present temperature-dependent EHC in Fig. 4 once GMs are adsorbed on all atoms. It is observed that the Schottky anomaly with both NO and NO₂ remains the same, while it takes place at about 2.5 eV with both NH₃ and CO. In addition, the EHC intensity decreases (increases) with NO and NO₂ (NH₃ and CO). In contrast to all observed phenomena so far, in the case of adsorption of CO on all atoms, there is a special range for EHC with strong responses from $k_B T \approx 1$ eV to 2.5 eV. This can be understood from the low-temperature behavior of CO on $\{b/d \text{ and } c\}$ atoms, see Fig. 2(b). The probability of transitions at low temperatures for CO and NH₃ is higher than NO and NO₂. As a result of this, one should expect a fingerprint of this behavior in the responses when these molecules are adsorbed on all atoms (stronger for CO but weaker for NH₃).

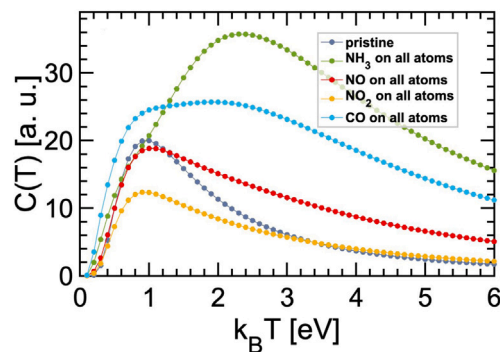


Fig. 4. The same as Fig. 2 but for the adsorption of GMs on all atoms. The Schottky anomaly is shifted with the adsorption of GM on all atoms, while the EHC intensity is increased (decreased) with the adsorption of NH₃/CO (NO₂/NO).

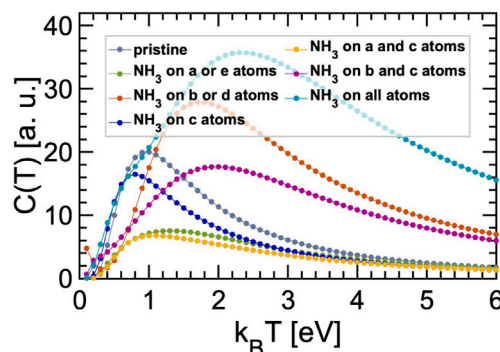


Fig. 5. The evolution of electronic heat capacity of β_{12} -borophene with the adsorption of NH₃ on different configurations of boron atoms.

To compare the results of the GMs for different configurations in the adsorption process, in Fig. 5, we explore the evolution of EHC and the Schottky anomaly with the GM, e.g. NH₃, adsorbed on different atoms. It can be seen that the intensity of EHC is strongly enhanced if the GM is adsorbed on all atoms. Also, due to the specific behavior of the adsorption process on b/d atoms, low-temperature responses are different resulting in the second dominant EHC intensity. For other configurations, we observe a suppression of EHC intensity except $\{b/d \text{ and } c\}$ atoms for the same reason above-mentioned. As for the evolution of thermal Schottky anomaly with GMs, adsorption on a/e , $\{a/e \text{ and } c\}$, and c atoms, does not significantly shift the critical thermal energy, while it does for the adsorption on b/d , $\{b/d \text{ and } c\}$, and all atoms.

5. Summary

In conclusion, the adsorption effect of gas molecules NH₃, NO, NO₂, and CO on the electronic heat capacity (EHC) and thermal Schottky anomaly of β_{12} -borophene is analyzed by the tight-binding model and the Green's function approach. We consider the inversion symmetry between boron atoms. We find that the EHC is largely enhanced (suppressed) for the adsorption on individual/all (pair) atoms. The dominant effect of gas molecules on the enhancement of the EHC is dedicated to the absorption of NO/NH₃ (NO₂) on individual/all (pair) atoms. On the other hand, the pristine Schottky anomaly shifts to the higher thermal energies with the adsorbed gas molecules. Overall, we provide the detail of host-guest couplings – gas molecules-boron atoms – for practical applications in field-effect transistor-based gas sensors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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